

Sharp edge bounds for bipartite graphs with unique minimum domination

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Revised version

Abstract

A graph with no isolated vertices and a unique minimum dominating set of cardinality γ has at least 3γ vertices. We determine the maximum number of edges in bipartite graphs at the next two orders. For every $\gamma \geq 2$, the maxima at orders $3\gamma + 1$ and $3\gamma + 2$ are respectively $\gamma(\gamma + 7)/2$ and $\lceil \gamma^2/2 \rceil + 5\gamma$; connected graphs attain both. At order $3\gamma + 1$, the extremal graph is unique up to isomorphism for every $\gamma \geq 2$. At order $3\gamma + 2$, at least two nonisomorphic extremal graphs exist for every even $\gamma \geq 4$. The results recover previously reported counterexamples to a conjecture of Koch and Narayan and place them in sharp general boundary theorems. The proofs are elementary: exterior private neighbours reduce the problem to local replacements on six, seven and eight vertices, with the domination obligations of the residual vertices retained explicitly.

Keywords: unique minimum dominating set; bipartite graph; extremal size; private neighbour.

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1 Introduction

All graphs in this paper are finite and simple. A set $D \subseteq V(G)$ is *dominating* if every vertex outside D has a neighbour in D . The domination number $\gamma(G)$ is the minimum cardinality of a dominating set. A graph has a *unique minimum dominating set* if exactly one dominating set has cardinality $\gamma(G)$. Throughout, uniqueness refers to minimum cardinality, not to inclusion-minimality. Write $n(G) = |V(G)|$ and $e(G) = |E(G)|$.

Fischermann, Rautenbach and Volkmann [1] investigated the maximum size of graphs with a unique minimum dominating set. Fraboni and Shank [2] subsequently settled their conjectured size bound when the domination number is two. Koch and Narayan [4] considered the bipartite problem and proposed a bound depending on the order and the domination number. They proved their bound for domination number two and at order $n = 3\gamma$. Erlbacher's public artifacts [3] already give the 13-vertex, 22-edge counterexample, an unbalanced family containing a 14-vertex, 28-edge example, and finite extremal computations. The results below recover those examples and determine the sharp maxima at the next two orders for general domination number. Koch and Narayan also identify the unique extremal graph at $(n, \gamma) = (10, 3)$ in their Figure 1 discussion.

The order restriction is intrinsic. If D is the unique minimum dominating set of a graph with no isolated vertices, every vertex of D has at least two exterior private neighbours with respect to D . These neighbours are distinct for distinct members of D , and hence $n(G) \geq 3\gamma(G)$. We recall the short argument in Lemma 3.1.

For $r \in \{1, 2\}$, let $\mathcal{B}_{\gamma,r}$ be the class of bipartite graphs with no isolated vertices, of order $3\gamma + r$, whose minimum dominating set is unique and has cardinality γ . Write $\mathcal{B}_\gamma = \mathcal{B}_{\gamma,1}$. No connectedness assumption is imposed in these definitions.

Theorem 1.1. *For every integer $\gamma \geq 2$,*

$$\max_{G \in \mathcal{B}_\gamma} e(G) = \frac{\gamma(\gamma + 7)}{2}.$$

The maximum is attained by a connected graph. For every $\gamma \geq 2$, equality holds if and only if

$$G \cong H(\lceil \gamma/2 \rceil, \lfloor \gamma/2 \rfloor),$$

where $H(p, q)$ is the graph defined in Section 2.

Theorem 1.2. *For every integer $\gamma \geq 2$,*

$$\max_{G \in \mathcal{B}_{\gamma,2}} e(G) = \left\lceil \frac{\gamma^2}{2} \right\rceil + 5\gamma.$$

The connected graph $H_2(\lceil \gamma/2 \rceil, \lfloor \gamma/2 \rfloor)$ attains the maximum. For every even $\gamma \geq 4$, at least two nonisomorphic connected graphs attain it. No complete classification of these equality cases is asserted.

At order $3\gamma + 1$, Conjecture 1 of [4] specializes to

$$e(G) \leq 2\gamma + 2ab + 2a + 1, \quad a = \lceil \gamma/2 \rceil, \quad b = \lfloor \gamma/2 \rfloor. \quad (1)$$

The summation term in the conjecture is empty at this order. Since

$$\frac{\gamma(\gamma + 7)}{2} - (2\gamma + 2ab + 2a + 1) = b - 1, \quad (2)$$

Theorem 1.1 has the following consequence.

Corollary 1.3. *For every $\gamma \geq 4$, there is a connected bipartite graph on $3\gamma + 1$ vertices with a unique minimum dominating set of cardinality γ that violates (1). The excess is $\lfloor \gamma/2 \rfloor - 1$. In particular, at $\gamma = 4$ the extremal graph has 13 vertices and 22 edges, whereas (1) gives 21.*

The additional vertex explains the change from the threshold case $n = 3\gamma$. It may be adjacent to several members of the unique minimum dominating set. Our extremal graph uses this possibility. The proof of the upper bound keeps track of whether the additional vertex has one or several such neighbours; the equality argument then forces the whole construction.

2 The extremal constructions

Definition 2.1. For integers $p, q \geq 1$ and $r \in \{1, 2\}$, let $H_r(p, q)$ have pairwise distinct vertices

$$X = \{x_i : 1 \leq i \leq p\}, \quad Y = \{y_j : 1 \leq j \leq q\},$$

$$U_i = \{u_i^0, u_i^1\}, \quad V_j = \{v_j^0, v_j^1\}, \quad Z = \{z_1, \dots, z_r\}.$$

Its edges are exactly $x_i u_i^\varepsilon$, $y_j v_j^\varepsilon$, $u_i^\varepsilon v_j^0$, $z_t u_i^\varepsilon$ and $z_t y_j$, for every admissible index and $\varepsilon \in \{0, 1\}$. Write $H(p, q) = H_1(p, q)$, and write z for its single member of Z .

The graph is bipartite, with parts

$$X \cup \bigcup_j V_j \cup Z \quad \text{and} \quad Y \cup \bigcup_i U_i. \quad (3)$$

It is connected: each z_t meets all of $Y \cup \bigcup_i U_i$, each x_i meets U_i , and each member of V_j meets y_j .

Proposition 2.2. *The graph $H_r(p, q)$ has order $3(p + q) + r$, domination number $p + q$, and unique minimum dominating set $X \cup Y$. Its size is*

$$e(H_r(p, q)) = 2(p + q) + 2pq + r(2p + q). \quad (4)$$

Proof. The vertex and edge counts follow from the definition. Clearly $D = X \cup Y$ dominates. Every dominating set K meets each closed neighbourhood $N[x_i] = \{x_i, u_i^0, u_i^1\}$ and each $N[v_j^1] = \{y_j, v_j^1\}$. These $p + q$ sets are pairwise disjoint, so $|K| \geq p + q$.

If $|K| = p + q$, it chooses exactly one member of each of these sets and nothing else. Thus every member of Z , and every v_j^0 , is absent. Choosing u_i^1 instead of x_i leaves its mate undominated: the mate's neighbours are x_i , the members of Z , and the vertices v_j^0 . Hence all x_i are selected and all U_i vertices are absent. Choosing v_j^1 instead of y_j now leaves v_j^0 undominated, forcing all y_j . Thus $K = D$. $\square$

For $a = \lceil \gamma/2 \rceil$ and $b = \lfloor \gamma/2 \rfloor$, direct calculation gives

$$e(H(a, b)) = 2ab + 4a + 3b = \frac{\gamma(\gamma + 7)}{2}, \quad (5)$$

$$e(H_2(a, b)) = 2ab + 6a + 4b = \left\lceil \frac{\gamma^2}{2} \right\rceil + 5\gamma. \quad (6)$$

These graphs supply the connected lower bounds for both theorems.

The subfamily $H_2(p, 1)$ is the two-extra-vertex case of Erlbacher's unbalanced-split construction [3], after interchanging the names of the center sides. In particular, $H_2(3, 1)$ is his 14-vertex example. For $q > 1$, each member of Z meets several centers; this shared-center feature already occurs in his 13-vertex counterexample, which is $H(2, 2)$. We use these constructions to establish sharpness, without claiming priority for those previously reported examples or mechanisms.

3 Private neighbours and local replacements

For a dominating set D and $d \in D$, define

$$\text{epn}(d, D) = \{u \in V(G) \setminus D : N(u) \cap D = \{d\}\}.$$

We use closed neighbourhoods only when square brackets are displayed.

Lemma 3.1. *If G has no isolated vertices and D is its unique minimum dominating set, then $|\text{epn}(d, D)| \geq 2$ for every $d \in D$.*

Proof. If $\text{epn}(d, D) = \{u\}$, the set $(D \setminus \{d\}) \cup \{u\}$ dominates G : u dominates itself and d , while every other vertex outside D retains a neighbour in $D \setminus \{d\}$. This contradicts uniqueness.

If $\text{epn}(d, D) = \emptyset$ and d has a neighbour in D , then $D \setminus \{d\}$ is a smaller dominating set. Otherwise d has no neighbour in D . The no-isolate assumption gives a neighbour u outside D , and $(D \setminus \{d\}) \cup \{u\}$ is again a different dominating set of the same cardinality. All cases are impossible. $\square$

The local lemmas below include every possible edge between the two centers. Their use is governed by the following replacement principle. If an induced subgraph F contains $x, y \in D$ and all other members of D lie outside F , and $D \setminus \{x, y\}$ dominates every vertex outside F , then any dominating set S of F of size at most two extends to $(D \setminus \{x, y\}) \cup S$. Thus, whenever $\{x, y\}$ dominates F , it is its unique dominating set of size at most two. In each application we verify the outside-domination condition explicitly; cross-cell edges cannot obstruct the resulting domination. The six-vertex argument is the local estimate used in the proof of Theorem 12 of [4]; we record its equality patterns as well.

Lemma 3.2. *Let F be a bipartite graph with parts $\{x\} \cup V$ and $\{y\} \cup U$, where $|U| = |V| = 2$. Assume that x is adjacent to both vertices of U and y to both vertices of V . If $\{x, y\}$ is the unique dominating set of cardinality at most two, then at most two edges occur among xy and the U – V edges. If exactly two occur, either xy is absent and the U – V edges form a two-edge star, or xy is present and there is exactly one U – V edge.*

Proof. Two disjoint U – V edges give an alternative dominating pair: select opposite endpoints, one from each edge. If xy is absent, three U – V edges contain two disjoint edges. Thus there are at most two, and a two-edge configuration must be a star.

Suppose xy is present. Two U – V edges either are disjoint or form a star. A star centered at $u \in U$ gives the dominating pair $\{x, u\}$; a star centered at $v \in V$ gives $\{y, v\}$. Consequently at most one U – V edge can accompany xy . $\square$

Lemma 3.3. *Let F have the same form as in Lemma 3.2, with $|U| = 2$ and $|V| = s \in \{3, 4\}$. If $\{x, y\}$ is its unique dominating set of size at most two, then at most $2s - 2$ optional edges occur among xy and U – V .*

Proof. If xy is absent and there are at least $2s - 1$ cross-edges, one U row is full and the other has a neighbour v . The full row vertex u and v form an alternative dominating pair. If xy is present, a V vertex adjacent to both members of U gives the alternative $\{y, v\}$. Hence every column has at most one cross-edge, leaving at most $s + 1 \leq 2s - 2$ optional edges in total. $\square$

Lemma 3.4. *In Lemma 3.3 with $s = 3$, equality holds exactly when xy is absent and the cross-edges form a $K_{2,2}$ on U and two members of V , leaving the third column empty.*

Proof. If xy is present at equality, there are three cross-edges and no full column. Each column therefore contains one edge. If one row contains all three, $\{x, u\}$ dominates. Otherwise the row degrees are two and one; the degree-two row vertex together with the other row's neighbour dominates. Both contradict uniqueness.

With xy absent, four cross-edges have row degrees $(3, 1)$ or $(2, 2)$. A full row vertex and a neighbour of the other row rule out $(3, 1)$. For $(2, 2)$ with different row neighbourhoods, select u_0 and the unique member of $N(u_1) \cap V$ outside $N(u_0)$. They dominate the whole cell. Thus both rows meet the same two columns.

Conversely, let c be the empty column. A dominating pair must meet both disjoint sets $N[x] = \{x, u_0, u_1\}$ and $N[c] = \{c, y\}$, and selects nothing else. Choosing a U vertex leaves its mate undominated, forcing x ; then choosing c leaves the active columns undominated, forcing y . No singleton can meet both disjoint sets. $\square$

Lemma 3.5. *Let F have the same form, now with $|U| = |V| = 3$. If $\{x, y\}$ is its unique dominating set of size at most two, then at most six optional edges occur.*

Proof. Suppose at least seven optional edges occur. If xy is absent, the 3 by 3 cross-edge matrix has at most two missing entries. A full row and a full column therefore exist; their vertices form an alternative dominating pair. If xy is present, there are at least six cross-edges. A full row gives $\{x, u\}$, and a full column gives $\{y, v\}$. Otherwise every row and column has degree at most two. Six edges force every degree to equal two, so the missing entries form a perfect matching. The endpoints of any missing edge dominate both centers and every private vertex, again a contradiction. $\square$

These elementary arguments give the optional-edge bounds 2, 4, 6, 6 for private-set sizes $(2, 2)$, $(2, 3)$, $(2, 4)$, $(3, 3)$ respectively. The exact checks in Appendix A are supplementary.

4 One residual vertex

Let $G \in \mathcal{B}_\gamma$, and let D be its unique minimum dominating set. By Lemma 3.1, we may select two exterior private neighbours for each vertex of D . The selected pairs are disjoint and account for 2γ vertices. Since $n(G) = 3\gamma + 1$, exactly one further vertex z remains.

Choose the names of the bipartition classes so that $z \in A$, and write

$$D \cap A = \{x_1, \dots, x_p\}, \quad D \cap B = \{y_1, \dots, y_q\}, \quad p + q = \gamma.$$

Let $U_i \subseteq B$ be the selected private pair of x_i , and $V_j \subseteq A$ the selected private pair of y_j . Set

$$J = N(z) \cap D \subseteq \{y_1, \dots, y_q\}, \quad t = |J|.$$

Because D dominates z , we have $t \geq 1$, and in particular $q \geq 1$.

Privacy and bipartiteness give a complete list of possible edges:

1. the 2γ selected center-to-private edges;
2. $x_i y_j$ and the edges between U_i and V_j , for each pair (i, j) ;
3. the edges from z to J and to $\bigcup_i U_i$.

In particular, edges inside D have not been discarded: they are precisely the possible edges $x_i y_j$ in the second class.

4.1 The additional vertex has several dominators

Suppose $t \geq 2$. For each i, j , consider the induced six-vertex graph

$$G[K_{ij}], \quad K_{ij} = \{x_i, y_j\} \cup U_i \cup V_j.$$

The set $D \setminus \{x_i, y_j\}$ dominates every vertex outside K_{ij} . Each selected private vertex outside the cell retains its own center; the vertex z retains a neighbour in J because only one y -center has been removed.

If $G[K_{ij}]$ had a dominating set S of cardinality at most two other than $\{x_i, y_j\}$, then

$$(D \setminus \{x_i, y_j\}) \cup S$$

would be a dominating set of G of cardinality at most γ , distinct from D when its cardinality is γ . This contradicts minimum cardinality or uniqueness. Thus Lemma 3.2 applies to every cell. The optional edge sets of these cells are disjoint, and $\deg(z) \leq 2p + q$. Consequently

$$e(G) \leq 2\gamma + 2pq + 2p + q. \tag{7}$$

When $p = 0$, the same bound follows directly from the edge list, with an empty sum over cells.

4.2 The additional vertex has one dominator

Suppose $J = \{y_{j_0}\}$. For every $j \neq j_0$, the preceding six-vertex replacement still applies, since y_{j_0} is retained and dominates z . For each i consider instead

$$L_i = \{x_i, y_{j_0}\} \cup U_i \cup V_{j_0} \cup \{z\}.$$

In $G[L_i]$, the center y_{j_0} has the three private vertices $V_{j_0} \cup \{z\}$. Every vertex outside L_i is dominated by $D \setminus \{x_i, y_{j_0}\}$, so the same replacement argument proves that Lemma 3.3 applies.

The optional edges of $G[L_i]$ are $x_i y_{j_0}$ and the edges from U_i to $V_{j_0} \cup \{z\}$. These edge sets are disjoint for different i . The edge $z y_{j_0}$ is counted once separately. Therefore

$$\begin{aligned} e(G) &\leq 2\gamma + 1 + 2p(q-1) + 4p \\ &= 2\gamma + 1 + 2pq + 2p \\ &\leq 2\gamma + 2pq + 2p + q. \end{aligned} \tag{8}$$

If $p = 0$, only the 2γ fixed private edges and $z y_{j_0}$ occur, so the displayed inequalities remain valid.

4.3 Optimizing the bipartition

Put $d = p - q$. Since $p + q = \gamma$, both cases yield

$$e(G) \leq 2\gamma + 2pq + 2p + q = \frac{\gamma(\gamma+7)}{2} - \frac{d(d-1)}{2} \leq \frac{\gamma(\gamma+7)}{2}. \tag{9}$$

The final inequality holds because d is an integer. Equality in the numerical optimization requires $d \in \{0, 1\}$. The parity of γ then forces

$$p = \lceil \gamma/2 \rceil, \quad q = \lfloor \gamma/2 \rfloor. \tag{10}$$

Together with (5), this proves the extremal-value assertion of Theorem 1.1 for every $\gamma \geq 2$.

5 Equality with one residual vertex

5.1 Domination number at least four

Suppose now that $\gamma \geq 4$ and $e(G) = \gamma(\gamma+7)/2$. Equation (10) gives $q = \lfloor \gamma/2 \rfloor \geq 2$. The final inequality in (8) loses $q-1$ edges, so the case $t=1$ cannot give equality. Every inequality in (7) must therefore be an equality. It follows that z is adjacent to every y_j and to every vertex of $\bigcup_i U_i$, and each six-vertex cell has exactly two optional edges.

We first show that D is independent. If $x_i y_j$ were an edge, then

$$(D \setminus \{x_i\}) \cup \{z\}$$

would be a different dominating set of cardinality γ . Indeed, x_i is dominated by the retained y_j , the pair U_i is dominated by z , and every other private pair retains its center. Thus no edge $x_i y_j$ occurs.

Lemma 3.2 now shows that the two edges between U_i and V_j form a star. This star cannot be centered at $u \in U_i$, because

$$(D \setminus \{x_i, y_j\}) \cup \{u, z\}$$

would dominate G . The vertex u dominates x_i and both members of V_j , while z dominates y_j and both members of U_i . All other private pairs retain their centers. Therefore both members of U_i are adjacent to exactly one vertex of V_j ; denote it by $v_j^{(i)}$.

Fix j . Suppose that the selected vertex $v_j^{(i)}$ is not constant as i varies. Choose one vertex $u_i \in U_i$ for every i , and define

$$K = \{z\} \cup \{u_i : 1 \leq i \leq p\} \cup ((D \cap B) \setminus \{y_j\}).$$

Then $|K| = 1 + p + (q-1) = \gamma$. Every x_i is dominated by u_i ; every vertex in $\bigcup_i U_i$ and every y -center is dominated by z . The two vertices of V_j are dominated because both possible choices of $v_j^{(i)}$ occur. Every other pair V_ℓ retains y_ℓ . Hence K is a second minimum dominating set, a contradiction.

For each j , rename the members of V_j so that their common selected vertex is $v_j^{(0)}$. The complete edge list is now exactly that of $H(p, q)$. Equation (10) proves the equality assertion in Theorem 1.1 for $\gamma \geq 4$.

5.2 Domination numbers two and three

At equality, (10) gives $q = 1$ and $p = 1$ or 2 . Write y for the sole Y -center and put $W = V_1 \cup \{z\}$. All three vertices of W meet y . The graph has $2p + 3$ fixed center/private edges, and its optional edges partition into the p blocks $x_i y$ and $U_i - W$. Each seven-vertex cell is legitimate for replacement: every outside vertex retains its own center, while z is inside the cell. Equality forces four optional edges in every block.

Lemma 3.4 forbids all center edges and makes each U_i complete to two of the three W columns. For $p = 1$ this is $H(1, 1)$. For $p = 2$, suppose the omitted columns differ. The active column pairs then cover W and have one common member w . Choose $u_i \in U_i$. The set $\{u_1, u_2, w\}$ dominates: the u_i cover x_i and all columns between them, while w covers both U pairs and y . This is a different dominating triple, a contradiction. The omitted columns therefore coincide, and the graph is $H(2, 1)$. This completes Theorem 1.1 for all $\gamma \geq 2$. As noted above, the latter finite equality graph was already identified in [4].

6 Two residual vertices

Let $G \in \mathcal{B}_{\gamma, 2}$ and let D be its unique minimum dominating set. Select two exterior private neighbours of each member of D , leaving two residual vertices z, w . Put

$$D \cap A = X = \{x_1, \dots, x_p\}, \quad D \cap B = Y = \{y_1, \dots, y_q\}, \quad p + q = \gamma,$$

and use $U_i \subseteq B$, $V_j \subseteq A$ for the selected private pairs. We prove the upper bound

$$M_2(\gamma) = \left\lceil \frac{\gamma^2}{2} \right\rceil + 5\gamma.$$

6.1 Residual vertices in the same part

Orient the parts so that $z, w \in A$. Each has a nonempty D -neighbourhood contained in Y , hence $q \geq 1$; $p = 0$ is allowed. Call a residual vertex *singleton-dominated* if it has one neighbour in D . Let $S \in \{0, 1, 2\}$ be the number of such vertices, and let r_j count those whose sole D -neighbour is y_j , so that $\sum_j r_j = S$.

For each i, j , take the cell with centers x_i, y_j , private set U_i , and opposite private set V_j together with the r_j residuals owned by y_j . Every vertex outside this cell remains dominated by $D \setminus \{x_i, y_j\}$: selected private vertices retain their centers; singleton residuals outside have a different retained owner; residuals with at least two D -neighbours retain one, because only one Y -center is removed. Lemmas 3.2 and 3.3 therefore bound its optional edges by $2 + 2r_j$.

These optional edge sets are pairwise disjoint. In addition to the 2γ fixed selected-private incidences, count the S singleton owner edges separately. Each of the remaining $2 - S$ residuals has degree at most $2p + q$: it can meet only Y and the U vertices. There is no edge zw . Consequently

$$\begin{aligned} e(G) &\leq 2\gamma + S + 2pq + 2pS + (2 - S)(2p + q) \\ &= 2\gamma + 2pq + 4p + 2q - S(q - 1) \\ &\leq 2pq + 6p + 4q. \end{aligned} \tag{11}$$

This includes $p = 0$, with no cells. For $d = p - q$, the final expression is

$$\frac{\gamma^2}{2} + 5\gamma + \frac{1}{2} - \frac{(d - 1)^2}{2}. \tag{12}$$

If γ is even, d is even and $(d - 1)^2 \geq 1$; if γ is odd, $(d - 1)^2 \geq 0$. Thus (12) is at most $M_2(\gamma)$.

6.2 Residual vertices in opposite parts

Suppose $z \in A$, $w \in B$, and write

$$J = N(z) \cap D \subseteq Y, \quad K = N(w) \cap D \subseteq X.$$

Both sets are nonempty, so $p, q \geq 1$. The edge zw is now possible and must be counted once.

Both residuals have several dominators. Suppose $|J|, |K| \geq 2$. Each ordinary six-vertex cell is valid: both residuals retain a D -neighbour after one X -center and one Y -center are removed. The cell edges total at most $2pq$. The edges incident with z other than zw number at most $2p + q$, and those incident with w other than zw number at most $2q + p$. Hence

$$e(G) \leq 2\gamma + 2pq + (2p + q) + (2q + p) + 1 = 2pq + 5\gamma + 1. \quad (13)$$

For odd γ , the inequality $2pq \leq \lfloor \gamma^2/2 \rfloor$ gives the result. For even γ , (13) is at most $M_2(\gamma) + 1$. An actual violation would, by integrality, force equality throughout, $p = q$, and saturation of both residual degree bounds. In particular, z meets every Y -center and every U vertex, while w meets every X -center and every V vertex. For any i, j , the set

$$(D \setminus \{x_i, y_j\}) \cup \{z, w\}$$

then dominates. The removed centers are covered by w, z respectively, U_i by z , and V_j by w ; all other private pairs retain their centers. This second dominating γ -set excludes the only possible excess.

Exactly one residual has one dominator. Interchange sides if necessary so that $J = \{y_{j_0}\}$ and $|K| \geq 2$. For $j \neq j_0$, use ordinary six-vertex cells. For $j = j_0$, include z and use a $(2, 3)$ private cell. Outside every such cell, w retains an X -neighbour; any residual not included keeps its owner. Thus the optional blocks contribute at most $2p(q - 1) + 4p$. Count zy_{j_0} once, the other incidences of w by $2q + p$, and zw at most once. This gives

$$e(G) \leq 2\gamma + 1 + 2p(q - 1) + 4p + (2q + p) + 1 = 2pq + 5\gamma + 2 - q. \quad (14)$$

If $q \geq 2$, this is at most $M_2(\gamma)$. If $q = 1$, then $p \geq 2$ because $|K| \geq 2$, so $\gamma \geq 3$. The difference between $M_2(\gamma)$ and the right side is $\lceil \gamma^2/2 \rceil - 2\gamma + 1$, which is zero at $\gamma = 3$ and positive for $\gamma \geq 4$.

Both residuals have one dominator. Write $J = \{y_{j_0}\}$ and $K = \{x_{i_0}\}$. Partition optional edges using $(2, 2)$ cells for $i \neq i_0, j \neq j_0$; $(3, 2)$ cells including w for $i = i_0, j \neq j_0$; $(2, 3)$ cells including z for $i \neq i_0, j = j_0$; and one $(3, 3)$ cell containing both residuals for $i = i_0, j = j_0$. Every residual whose owner is removed is included in the cell, and every residual outside retains its owner. All selected private vertices outside also retain their centers. Thus all local uniqueness hypotheses hold.

The edge zw belongs only to the $(3, 3)$ cell. The two residual owner edges are counted once each as fixed incidences, and the optional blocks are disjoint. The local bounds give

$$\begin{aligned} e(G) &\leq 2\gamma + 2 + 2(p - 1)(q - 1) + 4(q - 1) + 4(p - 1) + 6 \\ &= 2pq + 4\gamma + 2 \\ &\leq \lfloor \gamma^2/2 \rfloor + 4\gamma + 2 \leq M_2(\gamma), \end{aligned} \quad (15)$$

using $\gamma \geq 2$. This covers the small case $p = q = 1$ as well. Together with (6), all cases prove the extremal value in Theorem 1.2.

6.3 Nonisomorphic equality graphs for even domination number

Let $\gamma = 2k \geq 4$. Both $H_2(k, k)$ and $H_2(k+1, k-1)$ attain the maximum by (4). Their bipartition size pairs are respectively $\{3k, 3k+2\}$ and $\{3k+1, 3k+1\}$. A connected bipartite graph has its bipartition determined up to interchange, so the graphs are nonisomorphic. This proves the remaining assertion of Theorem 1.2.

7 Source comparison and further questions

At order $3\gamma + 1$, Theorem 1.1 exceeds the conjectured bound by $b - 1$, as in (2). At order $3\gamma + 2$, the same conjecture specializes to $2\gamma + 2ab + 4a + 2$. Its summation tail is again zero, both with the printed cutoff and its construction-consistent correction. Theorem 1.2 exceeds this expression by

$$(2\gamma + 2ab + 4a + 2b) - (2\gamma + 2ab + 4a + 2) = 2(b - 1).$$

Thus the two sharp boundary families violate the proposed bound for every $\gamma \geq 4$. The individual 13-vertex and 14-vertex counterexamples, and the cutoff issue, were already reported in [3]. Neither theorem claims a smallest counterexample over all orders.

For comparison, the published unbalanced-family expression in [3] at two extra vertices is $2\gamma + 2p'q' + 2(2q' + 1)$, with $p' + q' = \gamma$. When $p' = q, q' = p$, its count differs from (4) by $2(q - 1)$. At $\gamma = 5$, its maximum over positive splits is 36, whereas Theorem 1.2 gives 38, attained by $H_2(3, 2)$. This compares the explicit formulas, without asserting priority over every other possible construction.

The proofs give a complete equality classification at order $3\gamma + 1$. At order $3\gamma + 2$ we have established nonuniqueness for even $\gamma \geq 4$, but have not classified all extremal graphs. A further structural problem is to determine that equality class and to study fixed orders $3\gamma + r$ with $r \geq 3$. The present local argument does not by itself supply a general- r bound: the residual vertices' simultaneous domination obligations must be retained.

A Supplementary exact checks

All proofs above are elementary and independent of computer enumeration. The accompanying exact checks use only finite sets and integer bit masks. For a private cell with sizes (a, b) , fix the edges xU and yV and enumerate all subsets of the optional edges xy and $U \times V$. For each graph, test every vertex subset of size at most two. The resulting complete local counts are given in Table 1.

(a, b)	Optional patterns	Admissible patterns	Maximum optional edges
(2, 2)	32	14	2
(2, 3)	128	58	4
(2, 4)	512	254	6
(3, 3)	1024	548	6

Table 1: A pattern is admissible when its only dominating set of size at most two is $\{x, y\}$.

An independent implementation reproduces all 1696 local patterns. The low- γ equality reduction at order $3\gamma + 1$ was additionally checked on all 128 seven-vertex and all 16384 ten-vertex skeletons, including all subextremal edge patterns. Exactly three labelled equality patterns survive in each case, corresponding to the choice of the common omitted column. These are censuses of the proved reduced skeletons, not unrestricted graph censuses.

All vertex subsets were checked for the ten graphs $H_2(p, q)$ with $p, q \geq 1$ and $2 \leq p + q \leq 5$, of order at most 17; each has the claimed unique minimum dominating set. The balanced values are in Table 2. The one-residual family was also checked through $\gamma = 6$.

γ	(a, b)	$e(H(a, b))$	$e(H_2(a, b))$	Unique minimum sets
2	(1, 1)	9	12	1
3	(2, 1)	15	20	1
4	(2, 2)	22	28	1
5	(3, 2)	30	38	1

Table 2: The two boundary families at small domination numbers.

The 13-vertex graph was rechecked on all 8192 subsets and explicitly matched by an isomorphism to the prior certificate in [3]. An explicit bijection likewise matches all 28 edges of $H_2(3, 1)$ to that source’s 14-vertex certificate. Thus these checks certify recovery of the previously reported examples.

The proof package supplies the standalone checkers, exact JSON results, independent audit reports, and source hashes. The local checks and constructions are reproducible without a graph library or an optimization solver. They are not claimed as Lean formalizations or substitutes for the written proofs.

References

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